1.3 Exercises

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1

The zeroes of a function are all those x such that $f(x) = 0$.

2

A monomial an expression that is made of one term, that is, a variable, number, or a product of number and one or more variables. A binomial is an expression that is the sum two monomials, while a trinomial is an expression that is the sum of three monomials. An example of a monomial is $4x$, an example of a binomial is $4x + 3$, and an example of a trinomial is $y^2 + 4x + 3$.

3

$x^2 + 6x + 5$
$\rightarrow (x + 3)(x + 2)$

4

$x^2 - 7x + 10$
$\rightarrow (x - 5)(x - 2)$

5

$a^2 - 13a + 22$
$\rightarrow (a - 11)(a - 2)$

6

$r^2 + 15r + 56$
$\rightarrow (r + 8)(r + 7)$

7

Cannot be factored.

8

$q^2 - 11q + 28$
$\rightarrow (q - 7)(q - 4)$

9

$b^2 + 3b - 40$
$\rightarrow (b + 8)(b - 5)$

10

$x^2 - 4x - 12$
$\rightarrow (x - 6)(x + 2)$

11

$x^2 - 7x - 18$
$\rightarrow (x - 9)(x + 2)$

12

Cannot be factored.

13

$x^2 + 9x - 36$
$\rightarrow (x + 12)(x - 3)$

14

$m^2 + 8m - 65$
$\rightarrow (m + 13)(m - 5)$

15

$x^2 - 36$
$\rightarrow (x + 6)(x - 6)$

16

$b^2 - 81$
$\rightarrow (b + 9)(b - 9)$

17

$x^2 - 24x + 144$
$\rightarrow (x - 12)^2$

18

$t^2 - 16t + 64$
$\rightarrow (t - 8)^2$

19

$x^2 + 8x + 16$
$\rightarrow (x + 4)^2$

20

$c^2 + 28c + 196$
$\rightarrow (c + 14)^2$

21

$n^2 + 14n + 49$
$\rightarrow (n + 7)^2$

22

$s^2 - 26s + 169$
$\rightarrow (s - 13)^2$

23

$z^2 - 121$
$\rightarrow (z + 11)(z - 11)$

24

$x^2 - 8x + 12 = 0$
$\rightarrow (x - 6)(x - 2) = 0$
$\rightarrow x = 6 \vee x = 2$

25

$x^2 - 11x + 30 = 0$
$\rightarrow (x - 6)(x - 5) = 0$
$\rightarrow x = 6 \vee x = 5$

26

$x^2 + 2x - 35 = 0$
$\rightarrow (x + 7)(x - 5) = 0$
$\rightarrow x = -7 \vee x = 5$

27

$a^2 - 49 = 0$
$\rightarrow (a + 7)(a - 7) = 0$
$\rightarrow a = -7 \vee a = 7$

28

$b^2 - 6b + 9 = 0$
$\rightarrow (b-3)^2 = 0$
$\rightarrow b = 3$

29

$c^2 + 5c + 4 = 0$
$\rightarrow (c+4)(c+1) = 0$
$\rightarrow c = -4 \vee c = -1$

30

$n^2 - 6n = 0$
$\rightarrow n(n-6) = 0$
$\rightarrow (n-6)(n-0) = 0$
$\rightarrow n = 6 \vee n = 0$

31

$t^2 + 10t + 25 = 0$
$\rightarrow (t+5)^2 = 0$
$\rightarrow t = -5$

32

$w^2 - 16w + 48 = 0$
$\rightarrow (w-12)(w-4) = 0$
$\rightarrow w = 12 \vee w = 4$

33

$z^2 - 3z = 54$
$\rightarrow z^2 - 3z - 54 = 0$
$\rightarrow (z - 9)(z + 6) = 0$
$\rightarrow z = 9 \vee z = -6$

34

$r^2 + 2r = 80$
$\rightarrow r^2 + 2r - 80 = 0$
$\rightarrow (r + 10)(r - 8) = 0$
$\rightarrow r = -10 \vee r = 8$

35

$u^2 = -9u$
$\rightarrow u^2 + 9u = 0$
$\rightarrow u(9 + u) = 0$
$\rightarrow (9 + u)(u - 0) = 0$
$\rightarrow u = -9 \vee 9$

36

$m^2 = 7m$
$\rightarrow m^2 - 7m = 0$
$\rightarrow m(m - 7) = 0$
$\rightarrow (m - 7)(m - 0) = 0$
$\rightarrow m = 7 \vee m = 0$

37

$14x - 49 = x^2$
$\rightarrow x^2 - 14x + 49 = 0$

$\rightarrow (x-7)^2 = 0$
$\rightarrow x = 7$

38

$-3y + 28 = y^2$
$\rightarrow y^2 + 3y - 28 = 0$
$\rightarrow (y+7)(y-4) = 0$
$\rightarrow y = -7 \vee y = 4$

39

$3 - 2 = 1$, not -1, meaning that the correct factorization is $(x+2)(x-3)$, and hence, that $x = -2 \vee x = 3$

40

We need to write the function in intercept form in order to find the zeros; we should not solve the function to find values that equal 14. Instead:
$x^2 + 7x - 8 = 0$
$\rightarrow (x+8)(x-1) = 0$
$\rightarrow x = -8 \vee x = 1$

41

$x^2 + 2x - 63 = 0$
$\rightarrow (x+9)(x-7) = 0$
$\rightarrow x = -9 \vee x = 7$
Hence, the answers is A.

42

$2(24 \times 10) = (24 + x)(10 + x)$
$\rightarrow 480 = x^2 + 34x + 240$
$\rightarrow x^2 + 34x - 240 = 0$
$\rightarrow (x + 40)(x - 6) = 0$
$\rightarrow x = -40 \vee x = 6$. Becuase we can only use positive values, $x = 6$.

43

$3(10 \times 12) = (12 + x)(10 + x)$
$\rightarrow 360 = x^2 + 22x + 120$
$\rightarrow x^2 + 22x - 240 = 0$
$\rightarrow (x + 30)(x - 8) = 0$
$\rightarrow x = -30 \vee x = 8$. BEcause we can only use positive values, $x = 8$.

44

$x^2 + 6x + 8 = 0$
$\rightarrow (x + 4)(x + 2) = 0$
$\rightarrow x = -4 \vee x = -2$

45

$x^2 - 8x + 16 = 0$
$\rightarrow (x - 4)(x - 4) = 0$
$\rightarrow x = 4$

46

$x^2 - 4x - 32 = 0$
$\rightarrow (x - 8)(x + 4) = 0$
$\rightarrow x = 8 \vee x = -4$

47

$x^2 + 7x - 30 = 0$
$\rightarrow (x + 10)(x - 3) = 0$
$\rightarrow x = -10 \vee x = 3$

48

$x^2 + 11x = 0$
$\rightarrow x(x + 11) = 0$
$\rightarrow (x + 11)(x - 0) = 0$
$\rightarrow x = -11 \vee x = 0$

49

$x^2 - 8x = 0$
$\rightarrow (x - 0)(x - 8) = 0$
$\rightarrow x = 0 \vee x = 8$

50

$x^2 - 64 = 0$
$\rightarrow (x - 8)(x + 8) = 0$
$\rightarrow x = 8 \vee x = -8$

51

$x^2 - 25 = 0$
$\rightarrow (x - 5)(x + 5) = 0$
$\rightarrow x = 5 \vee x = -5$

52

$x^2 - 12x - 45 = 0$
$\rightarrow (x - 15)(x + 3) = 0$
$\rightarrow x = 15 \vee x = -3$

53

$x^2 + 19x + 84 = 0$
$\rightarrow (x + 12)(x + 7) = 0$
$\rightarrow x = -12 \vee x = -7$

54

$x^2 + 22x + 121 = 0$
$\rightarrow (x + 11)(x + 11) = 0$
$\rightarrow x = -11$

55

$x^2 + 2x + 1 = 0$
$\rightarrow (x + 1)(x + 1) = 0$
$\rightarrow x = -1$

56

$x^2 + 6x - 55 = 0$
$\rightarrow (x + 11)(x - 5) = 0$
$\rightarrow x = -11 \vee x = 5$ Hence, the answers is B.

57

$x^2 - 19x + 88$

58

For the expression $x^2 + bx + 7$ to be factored, $b = m + n$ where $mn = 7$. Hence, the only two such cases are for $m = 7, n = 1$, and $m = -7, n = -1$. Thus, the formula $x^2 + bx + 7$ can be factored iff $b = 8 \vee b = -8$.

59

$36 = (x + 5)x$
$\rightarrow x^2 + 5x - 36 = 0$
$\rightarrow (x + 9)(x - 4) = 0$
$\rightarrow x = -9 \vee x = 4$, and thus, because only positive values are acceptable, $x = 4$.

60

$84 = (x + 7)(x + 2)$
$\rightarrow x^2 + 9x + 14 = 84$
$\rightarrow x^2 + 9x - 70 = 0$
$\rightarrow (x + 14)(x - 5) = 0$
$\rightarrow x = 5 \vee x = -14$, and thus, because only positive values are acceptable, $x = 5$.

61

$42 = \frac{1}{2}(x + 3)(2x + 8)$
$\rightarrow \frac{1}{2}(2x^2 + 14x + 24) = 42$
$\rightarrow x^2 + 7x + 12 = 42$
$\rightarrow x^2 + 7x - 30 = 0$
$\rightarrow (x + 10)(x - 3) = 0$
$\rightarrow x = -10 \vee x = 3$, and thus, because negative values are ruled out, $x = 3$.

62

$32 = \frac{1}{2}(x + 6 + x + 2)x$
$\rightarrow \frac{1}{2}(2x^2 + 8x) = 32$
$\rightarrow x^2 + 4x = 32$
$\rightarrow x^2 + 4x - 32$
$\rightarrow (x + 8)(x - 4) = 0$
$\rightarrow x = -8 \vee x = 4$, and thus, because negative values are ruled out, $x = 4$.

63

We will choose the values 0 and 20, both 10 away from 10 on a number line: $x^2 - 20x$

64

(a):
For $x^2 + 16 = (x + m)(x + n)$ to be true, $x^2 + 16 = x^2 + x(m + n) + mn$ would be true. Now, these means that $m + n = 0$. For, if it were anything other than zero, then either $m+n = 16$ or $m+n \neq 16$. In the first case, we can simply pick any value other than 0 for x, and we recieve some $x(m + n) + mn \neq 16$, and in the second case, we can make $x = 0$, and thus $x(m+n)+mn \neq 16$ as $mn \neq 16$. Thus, we can conclude that $m + n = 0$. Hence, because $x(m + n) + mn = 16$, we can conclude that $mn = 16$. The only way these two equations can compute is if m and n had opposite signs, which is impossible, due to 16 being positive. Hence, no m and n exist that satisfy these conditions. Hence, we can conclude that, in general, $x^2 + k$ cannot be factored for any k (and this can easily be shown, by replacing 16 in the above by k).

65

$3(100 \times 50) = (100 + x)(50 + x)$
$\rightarrow 15000 = x^2 + 150x + 5000$
$\rightarrow x^2 + 150x - 1000 = 0$
$\rightarrow (x + 200)(x - 50) = 0$
$\rightarrow x = -200 \vee x = 50$, and thus the dimensions of the new park are 150 by 100 ft. wide.

66

$2(35 \times 18) = (35 + x)(18 + x)$
$\rightarrow 1260 = x^2 + 53x + 630$
$\rightarrow x^2 + 53x - 630 = 0$
$\rightarrow (x + 63)(x - 10) = 0$
$\rightarrow x = -63 \vee x = 10$, and thus, the new dimensions of the enclosure are 45 feet long by 28 feet wide.

67

(a): $A = 20 \times 30 = 600 ft.^2$
(b): $600 + 464 = (30 + x)(20 + x)$
(c): $1064 = x^2 + 50x + 600$
$\rightarrow x^2 + 50x - 464 = 0$
$\rightarrow (x + 58)(x - 8) = 0$
$\rightarrow x = -58 \vee x = 8$, and thus, the length and the width of the of the patio should be expanded by 8 ft.

68

(a): $x^2 + 5x + 6$
(b): $(x+3)(x+2)$ The model an x grouped with 2 ones, and an x grouped with 3 ones, which is similar to $x + 2$ and $x + 3$, the terms of the factorization.

69

$3(18 \times 15) = (18 + x)(15 + x)$
$\rightarrow 810 = x^2 + 33x + 270$
$\rightarrow x^2 + 33x - 540 = 0$
$\rightarrow (x + 45)(x - 12) = 0$
$\rightarrow x = -45 \vee x = 12$, and thus, the length of the new rope need is 30 ft., and the width of the new rope needed is 27 ft.

70

$\frac{21\times20}{2} = (21-x)(20-x)$
$\rightarrow 210 = x^2 - 41x + 420$
$\rightarrow x^2 - 41x + 210 = 0$
$\rightarrow (x-35)(x-6) = 0$
$\rightarrow x = 35 \vee x = 6$, because negative measurements are ruled out, $x = 6$. Thus, the new deck is 15 ft. by 14 ft.

71

$200 = (10+x)(10+x)$
$\rightarrow x^2 + 20x - 100 = 0$
$\rightarrow$ Because $x^2 + 20x - 100$ cannot be factored, the problem cannot be solved by factoring.

72

This problem can be modeled with an equation that increases the width and length of the old lot and grocery store both by x, and then subtracting the area of the grocery store from the total area obtained by solving the quadratic equation:
$(240+x)(375+x) = 130500$
$\rightarrow x^2 + 615x + 90000 = 130500$
$\rightarrow x^2 + 615 - 40500 = 0$
$\rightarrow (x+675)(x-60) = 0$
$\rightarrow x = -675 \vee x = 60$, and thus, because negative values are ruled out, $x = 60$.